

Excel Coding Challenge

E-Commerce Sales Analysis

Key Observations:

Dataset Size & Structure

- The dataset contains **51,290 rows** and **13 columns**.
- Columns include order details, customer information, product category, and financial metrics like **Sales**, **Quantity**, **Discount**, and **Profit**.

Missing Values Are Significant

Several columns have **large numbers of missing values**:

- **Order ID**: 3,295 missing
- **Customer Name**: 3,194 missing
- **Sales**: 3,134 missing
- **Discount**: 3,383 missing
- **Profit**: 3,404 missing, etc.

Like this there are blank cell in almost all the columns

Data Types Are Incorrect

Most numeric columns (**Sales**, **Quantity**, **Discount**, **Profit**) are stored as **object/text**, not numeric.

Categorical Columns Mostly Complete

- Columns like **Region**, **Category**, and **Sub-Category** have minimal missing values.
- **Sub-Category** is completely filled (0 missing), making it reliable for segmentation and analysis.

Data Entry & Formatting

- Convert all 3 tables into a proper table to filter the blank cells and to change all the missing values.
- No duplicates are found.

- Delete the Missing Order Id rows.

Many other fields with these missing Order IDs are also missing
Customer Name missing, Country missing, Sales missing, Profit missing,
Discount missing, Quantity missing

So the rows without Order ID are not just missing Order ID they are also missing many important columns.

- Change the Order date and Ship Date in correct Date format.
- Blank Customer Name is changed to Unknown Customer.
- Clean all the columns as per the data needs.
- The Sales, Discount, and profit blank cells are changed to the average of the particular columns.

Create Pivot Tables

Create multiple pivot tables for key analysis:

1. Sales & Profit by Category
2. Top Products
3. Sales by Region/Country
4. Monthly/Yearly Trend
5. Discount vs. Profit Analysis

Create Dashboard

1. Key Metrics Cards:

- Total Sales
- Total Profit
- Total Orders
- Growth %

2. Charts to Visualize Trends:

- **Line Chart:** Yearly/Monthly Sales Trend
- **Pie Chart:** Profit by Category
- **Bar Chart:** Top 10 Products

- **Clustered Column Line Chart:** Country-wise sales by Category

3. **Insights Box:**

- Last month growth is slightly negative at -3.96%.
- Office Supplies contribute the largest share of profit.

Interactivity

- Use **Slicers** for Category, Region, Month/Year to make the dashboard interactive.
- Link charts to pivot tables so they update when slicers change.